

1. Difference between ratio and Proportion

Aspect	Ratio	Proportion
Definition	A ratio is a relationship between two numbers showing how many times the first number contains the second.	Proportion is a statement that two ratios are equal.
Purpose	To compare two quantities or show the relationship between them.	To express that two ratios are equivalent.
Use	Used to compare quantities, express rates (ex: speed), and in financial applications (ex: interest rates).	Used when you want to establish that two ratios are equivalent and solve for unknown quantities.
Mathematical Relationship	A ratio can be expressed as a fraction or a decimal.	A proportion is an equation stating that two ratios are equal, and it is solved using cross-multiplication
Measurement	A ratio can compare like or unlike quantities, such as time, distance, or size.	Proportion is used when we want to maintain the equality of ratios, often to find an unknown quantity in a set of equivalent ratios.

2. Modified de facto method in Bangladesh

The De Facto Method refers to a method used in population censuses and surveys to count people based on their actual place of residence or location at the time of the survey, rather than their usual or permanent place of residence. In the context of Bangladesh, the Modified De Facto Method has been used in recent Population and Housing Censuses. This approach takes into account the people who are physically present in a particular area at the time of the census enumeration, regardless of where they usually reside.

➤ Modified De Facto Approach in Bangladesh:

- **Temporary Migration:** Bangladesh has a high rate of internal migration (especially from rural to urban areas), seasonal migration for work, and international migration. The modified method ensures that these migrant populations are properly counted, even if they do not live in the area permanently.
- **Refugees and Informal Settlements:** The Modified De Facto Method also helps in counting refugees or people living in informal settlements, who might not have a permanent address. For example, Rohingya refugees living in camps in Cox's Bazar are included in the census counts using the de facto method.
- **Urbanization:** Since Bangladesh is experiencing rapid urbanization, this method better captures populations in growing urban areas, including floating populations (ex: day laborers, street vendors)

➤ Specific Features of the Modified De Facto Method in Bangladesh:

- **Census Night Reference:** Enumerators count all people who are physically present in a specific area on the census night.
- **Verification of Usual Residence:** In some cases, a follow-up question is asked to verify whether the person usually resides in the area or has just temporarily arrived. If the person is a temporary migrant, they are still counted in the current location.
- **Data Quality Assurance:** Special attention is given to ensure that the data is as accurate as possible, considering the mobility of people and the varied living conditions, especially in rural-urban migration.

3. Census vs Survey difference

Aspect	Census	Survey
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Definition	A census is a complete count of all people or things in a population.	A survey collects data from a selected sample of people or items.
Scope	Covers the entire population.	Covers only a sample of the population.
Frequency	Conducted at regular intervals (ex: every 10 years).	Can be conducted regularly or occasionally.
Cost	Expensive due to its wide coverage and large-scale efforts.	Generally cheaper, as it focuses on a smaller sample.
Time	Takes longer to collect and process data.	Takes less time, as fewer people are involved.

4. Population science vs Demography difference

Aspect	Population Science	Demography
Definition	Population science is a broader field that studies population dynamics, including growth, distribution, and structure.	Demography is a subfield of population science focused specifically on the statistical study of human populations.
Scope	Encompasses a variety of factors like migration, birth rates, death rates, and social behaviours.	Primarily focuses on population statistics, trends, and structures.
Focus	Focuses on understanding and addressing issues related to population growth, sustainability, and policy.	Focuses on analysing population size, distribution, and composition.
Methods	Includes statistical analysis, modelling, and policy analysis.	Primarily uses statistical data, census records, and surveys.
Example Topics	Urbanisation, environmental sustainability, migration patterns.	Birth rates, death rates, fertility rates, life expectancy.

5. De facto vs De jure difference

Aspect	De Facto	De Jure
Definition	Refers to what actually happens in practice, regardless of the law.	Refers to what is legally recognized or established by law.
Meaning	"In fact" or "in practice."	"By law" or "according to law."
Application	Applies to situations where something exists or happens in reality, even if not officially recognized.	Applies to situations that are officially or legally sanctioned.
Example in Population	De facto population count includes anyone present in an area during a survey, regardless of their legal residence.	De jure population count only includes people who are legally registered or considered residents.
Stability	Can be temporary or fluid, depending on circumstances.	More stable and permanent, as it is based on laws and regulations.

6. Write the Limitations of census.

Here are some limitations of a census:

- **High Cost:** Conducting a census is expensive due to the extensive data collection, logistics, and manpower involved.
- **Time-Consuming:** The process of gathering, processing, and analyzing data can take a long time.
- **Inaccurate Data:** There may be errors in data collection, such as underreporting, overreporting, or misreporting.
- **Non-Response:** Some individuals may not respond to census questionnaires, leading to incomplete data.

- **Coverage Issues:** Hard-to-reach populations (ex: remote areas, homeless, refugees) might be missed.
- **Privacy Concerns:** Some individuals may refuse to participate due to concerns about privacy and data security.
- **Temporary Migration:** The census might fail to accurately account for temporary migrants, affecting population counts in certain areas.

7. Why do we need demographic data?

Here are the reasons why we need demographic data:

- **Policy Formulation:** Helps governments design effective policies related to healthcare, education, and social welfare based on population needs.
- **Resource Allocation:** Ensures fair distribution of resources like food, healthcare, and infrastructure by identifying population density and regional needs.
- **Public Health:** Helps in tracking disease patterns, mortality rates, and health trends, aiding in better public health strategies.
- **Education Planning:** Aids in determining the need for schools, teachers, and educational resources based on age demographics.
- **Urbanization and Infrastructure:** Assists in urban planning by identifying areas of rapid population growth, helping to plan roads, utilities, and housing.
- **Election Boundaries and Representation:** Demographic data is used to determine electoral boundaries and ensure fair political representation.
- **Disaster Management:** Supports disaster response and recovery by identifying vulnerable populations and regions affected by natural disasters.

8. Why do we need a sample survey?

- Cost-Effective.
- Time-Saving.
- Easier to Manage.
- Useful for Frequent Studies.
- Provides valuable data for planning.
- Accurate Results.
- Allows researchers to focus on specific groups.
- Supports Quick Decision Making.

9. Survey is better than Census, explain.

A survey is often better than a census in terms of cost, time, and efficiency. While a census aims to collect data from the entire population, it is expensive, time-consuming, and logistically complex. In contrast, a well-designed survey collects information from a representative sample, providing accurate and reliable results at a much lower cost and in less time. Surveys are more flexible, can be conducted more frequently, and are easier to manage, especially in large or rapidly changing populations. Therefore, for many research and planning purposes, surveys are a more practical and efficient option than a full census.

10. Difference between Ascribed and Achieved characteristics.

Aspect	Ascribed Characteristics	Achieved Characteristics
Definition	Traits or statuses a person is born with or assigned involuntarily.	Traits or statuses a person gains through effort or choice.
Based On	Birth, family background, or natural factors.	Personal actions, decisions, education, or achievements.
Control	Cannot be changed easily or controlled.	Can be changed or earned over time.
Examples	Gender, race, caste, ethnicity, family name.	Education level, job position, income, skills.

Change Over Time	Usually fixed throughout life.	Can change with effort and time.
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11. Difference between Sex and Gender.

Sex	Gender
Biological difference between male and female.	Social and cultural identity of a person.
Assigned at birth based on body parts.	Based on personal identity and society's roles.
Includes male, female, and intersex.	Includes man, woman, non-binary, etc.
Usually fixed and does not change easily.	Can change over time and is flexible.
Same meaning across all societies.	Varies between cultures and time periods.

12. What is the relation between demographic variables and demographic process?

Demographic variables and demographic processes are closely linked. Demographic variables are the basic characteristics of a population, such as age, sex, education, occupation, income, and marital status. These variables help us understand what the population looks like. Demographic processes are the events that cause changes in the population. The main processes are birth (fertility), death (mortality), and migration.

These two are closely related:

- Demographic variables affect the processes. For example, a young population may have higher birth rates. Educated people may have fewer children. Job seekers may migrate to cities.
- Demographic processes change the variables. For example, births increase the number of children (age variable), deaths change the age and sex structure, and migration changes the population size and mix.

So, demographic variables and processes are connected. Variables help explain why changes happen, and processes cause those changes in the population over time.

13. Write differences between economic growth VS economic development.

Economic Growth	Economic Development
Increase in a country's output or income (GDP).	Improvement in people's living standards and well-being.
Focuses only on quantity (more goods and services).	Focuses on both quantity and quality (health, education, etc.).
Measured by GDP or national income.	Measured by HDI (Human Development Index), literacy, life expectancy, etc.
Can happen without development.	Development includes growth along with social progress.
Short-term change is possible.	Long-term process that takes time.
May benefit only a few people.	Aims to benefit the whole population.